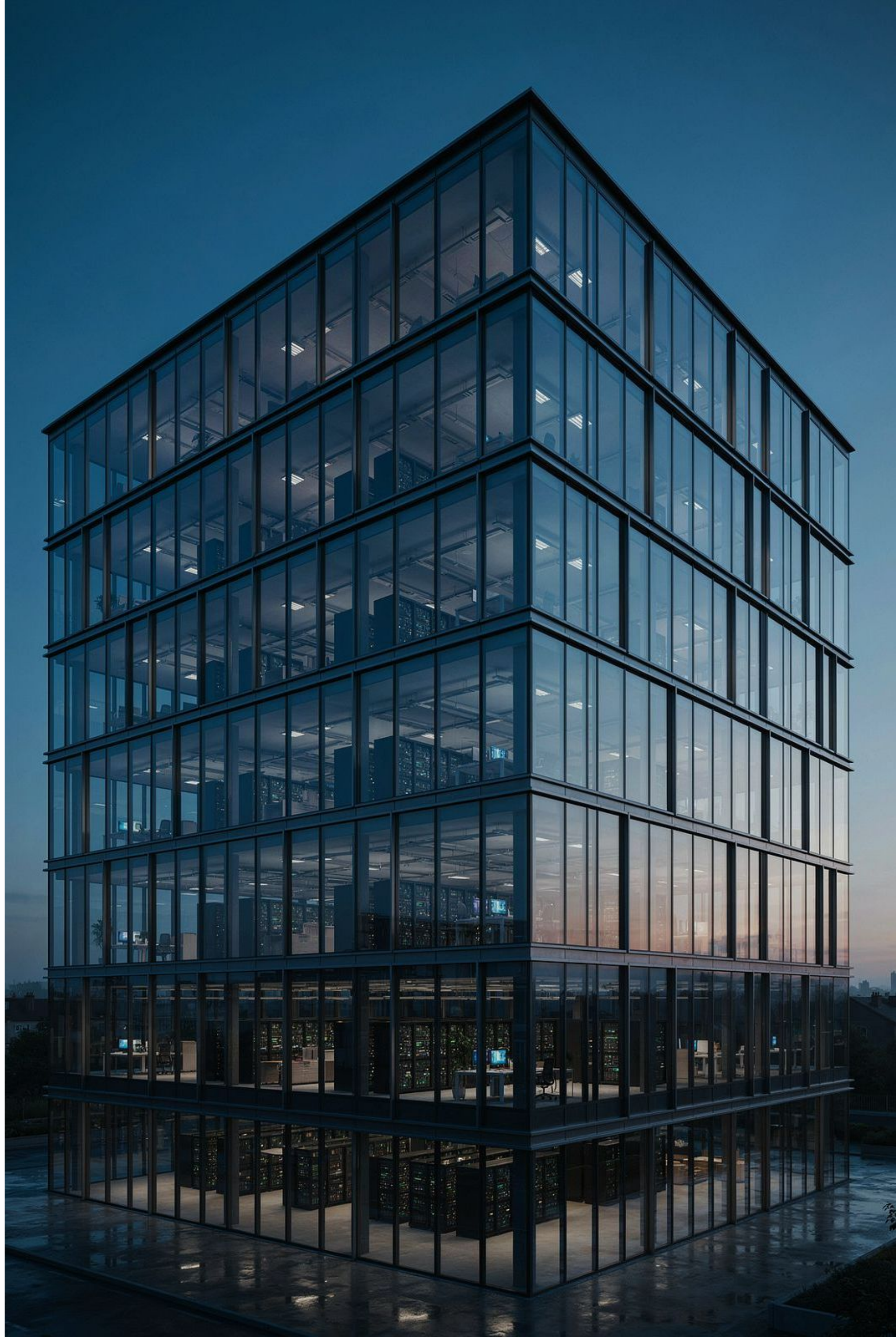
Day 1: IICS Architecture, Secure Agent & Connectors

Welcome to your first day with **Informatica Intelligent Cloud Services (IICS)** — also known as IDMC (Intelligent Data Management Cloud). Today we'll build a solid foundation: what IICS is, how it works under the hood, and how it connects to your real-world systems. By the end of the day, you'll understand the architecture, the agents, the connectors, and how to troubleshoot when things go wrong.

DAY 1 TRAINING

IT INTEGRATION ENGINEERS



Today's Agenda

Eight sessions, one clear goal: give you the mental model and hands-on confidence to work with IICS every day.

01

IICS / IDMC Overview

Platform overview, services, and terminology

02

IICS Architecture

Cloud service, Runtime, Secure Agent, job execution

03

Secure Agent Fundamentals

Groups, services, status, and configuration

04

Connections & Connectors

What they are, properties, testing, metadata

05

Common Enterprise Connectors

Flat File, DB, Salesforce, REST, SFTP, cloud storage

06

Connection Configuration

Credentials, ports, drivers, SSL, firewall, proxy

07

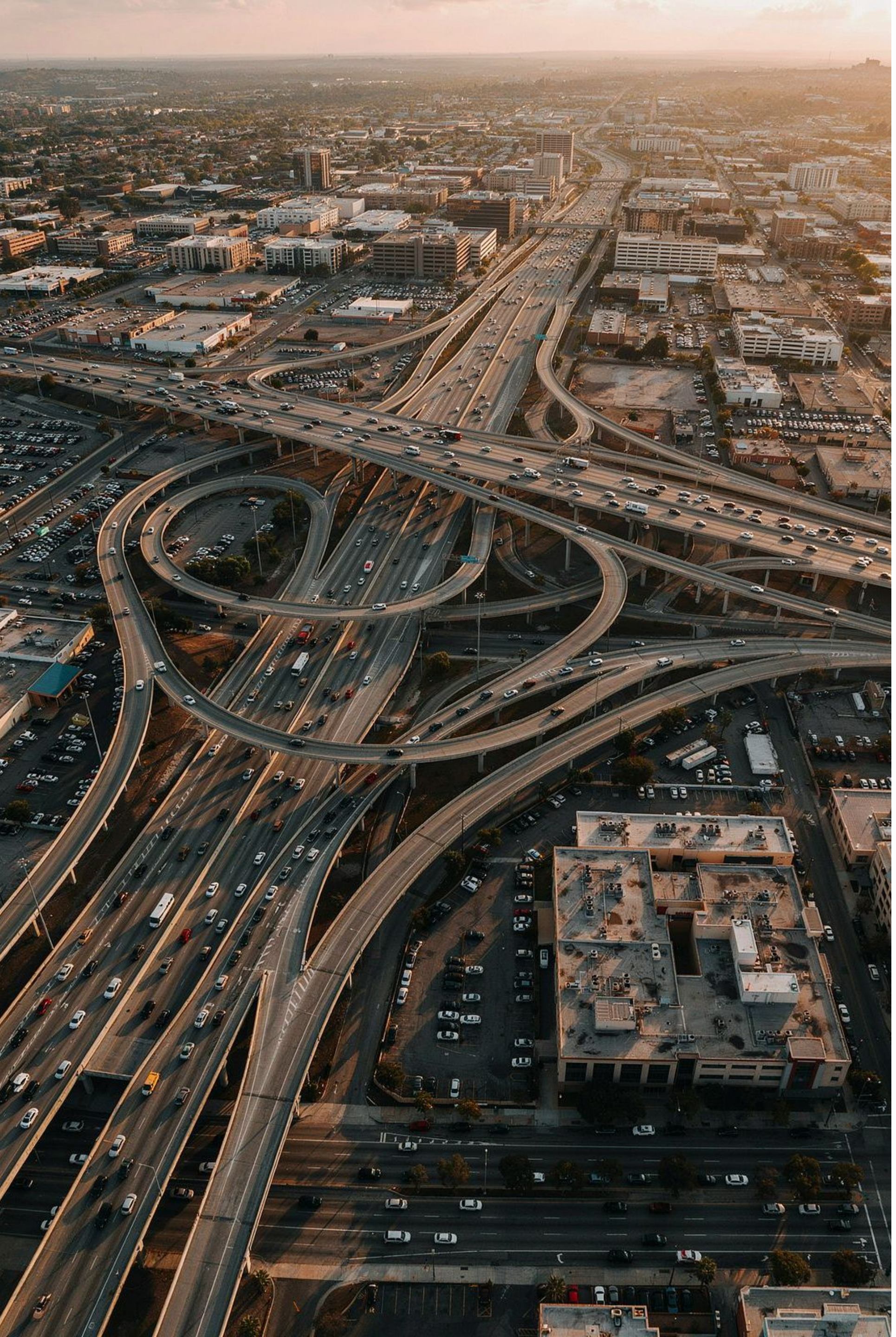
Metadata & Data Preview

Browsing objects, field metadata, validating access

08

Connector Troubleshooting

Failures, driver issues, logs, and fixes



Session 1: IICS / IDMC Overview

Before writing a single mapping, let's understand what IICS actually *is* — and why enterprises choose it.

What Is IICS? The "Airport" Analogy

Think of IICS as a **major international airport**. Data from many different sources (flights arriving) needs to land, get processed, and reach its destination (departing flights). IICS is the airport infrastructure — it doesn't store your data permanently, it *moves and transforms* it reliably between systems.

Cloud Data Integration

Build mappings and tasks that move and transform data between source and target systems — the core service you'll use daily.

Enterprise Use Cases

Sync Salesforce leads to your data warehouse. Load flat files into Oracle. Replicate databases. Automate nightly ETL jobs. These are all classic IICS scenarios.

IDMC Umbrella

IDMC is the broader Informatica platform. IICS (Cloud Data Integration) is one service within it. Other services include Data Quality, API management, and Master Data Management.

IICS Terminology You'll Hear Every Day

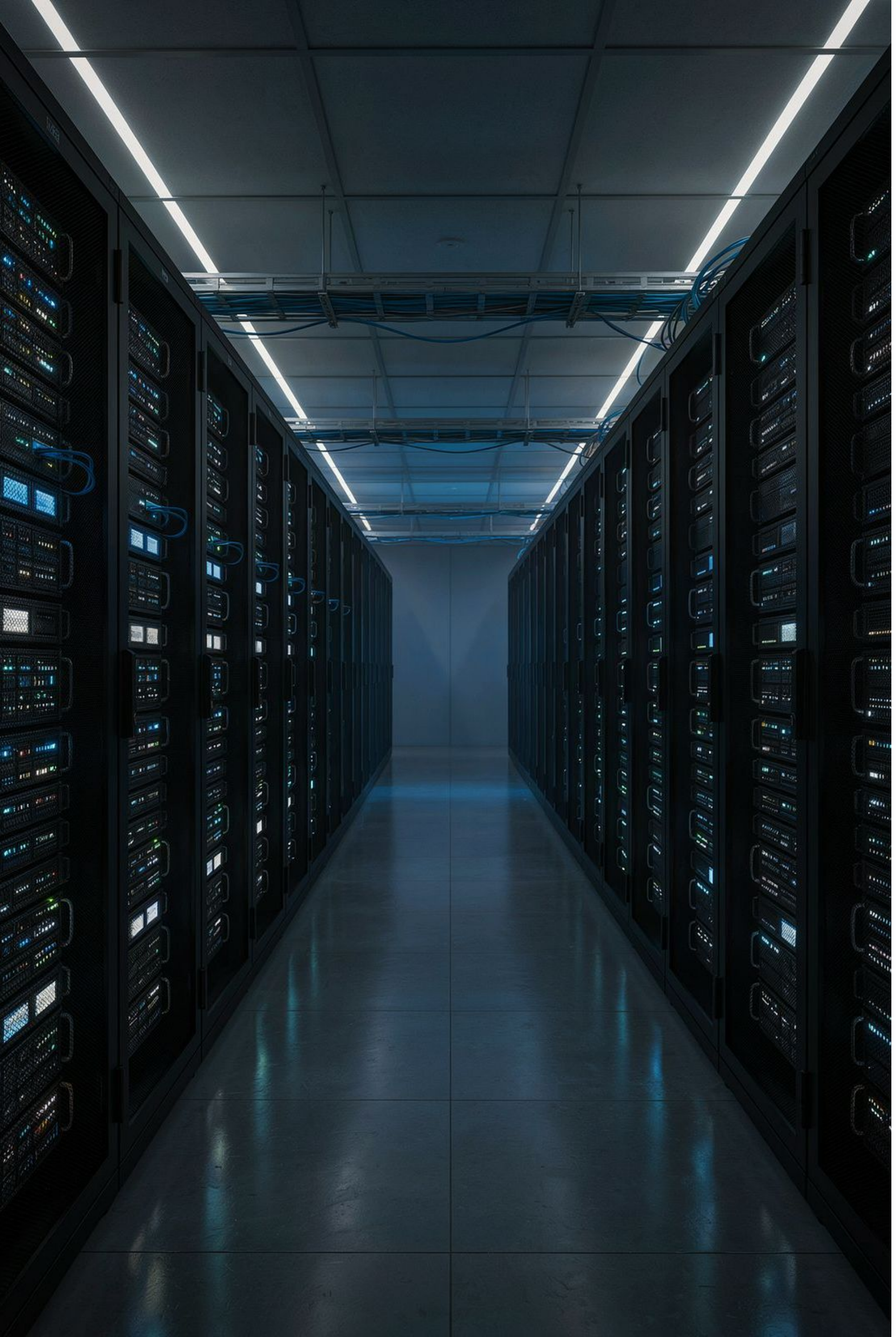
Like any platform, IICS has its own vocabulary. Here are the key terms — mapped to real-world concepts so they stick.

IICS Term

- **Organization** — Your tenant/account in IICS
- **Mapping** — The blueprint for moving/transforming data
- **Task** — A runnable instance of a mapping
- **Taskflow** — A sequence of tasks (like a workflow)
- **Runtime Environment** — Where tasks actually execute
- **Secure Agent** — Software installed on your network
- **Connection** — Credentials + settings for one system

Real-World Equivalent

- Your company's workspace or project folder
- A recipe — ingredients (sources) and steps (transformations)
- Baking the recipe once — a specific run
- A meal plan — many recipes in order
- The kitchen where cooking actually happens
- Your on-premises chef who has access to local ingredients
- The login card for one specific pantry or fridge



Session 2: IICS Architecture

Now that we know the vocabulary, let's understand **how the engine works** — from the cloud control plane down to the data flowing through your network.

Cloud Architecture: The Two Planes

IICS operates across two distinct planes. Understanding this split is the single most important architectural concept in IICS. Think of it like a **remote-controlled drone**: the controller (cloud) sends instructions, but the drone (Secure Agent) does the actual flying on-site.

☁️ Cloud Service (Control Plane)

Hosted by Informatica. This is where you **design, configure, and schedule** your integrations. Accessible via browser at **https://dm-ap.informaticacloud.com/identity-service/home**. No data lives here — only metadata and instructions.

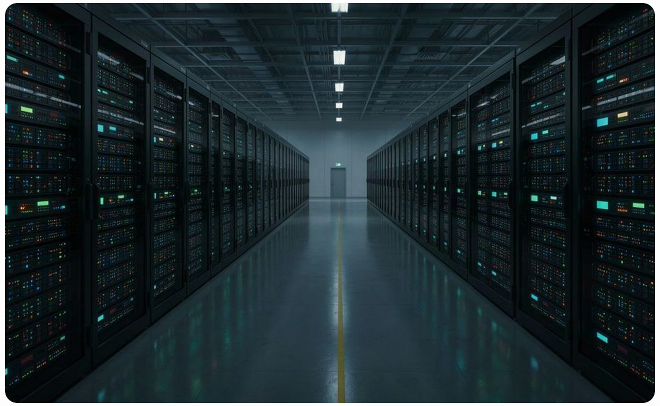
🖥️ Runtime Environment (Data Plane)

This is where data actually **moves and gets processed**. It runs either on a Secure Agent (your network) or on Informatica's cloud-hosted runtime. Your sensitive data never has to leave your premises if you use a Secure Agent.

Key rule: The cloud plane **orchestrates**. The runtime plane **executes**. They are always separated.

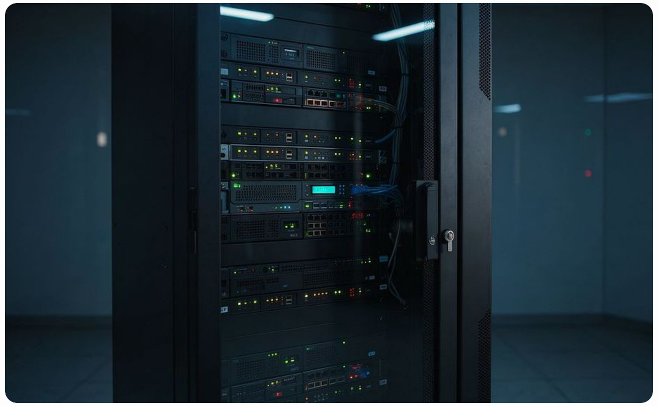
Cloud-Hosted vs. Secure Agent Execution

When you run a task, IICS must decide *where* to execute it. You choose based on where your data lives and what your security team allows.



Cloud-Hosted Runtime

Informatica runs the job on their shared infrastructure. Best for **cloud-to-cloud** integrations (e.g., Salesforce to Snowflake) where data doesn't need to touch your network. Fast to set up — no software to install.

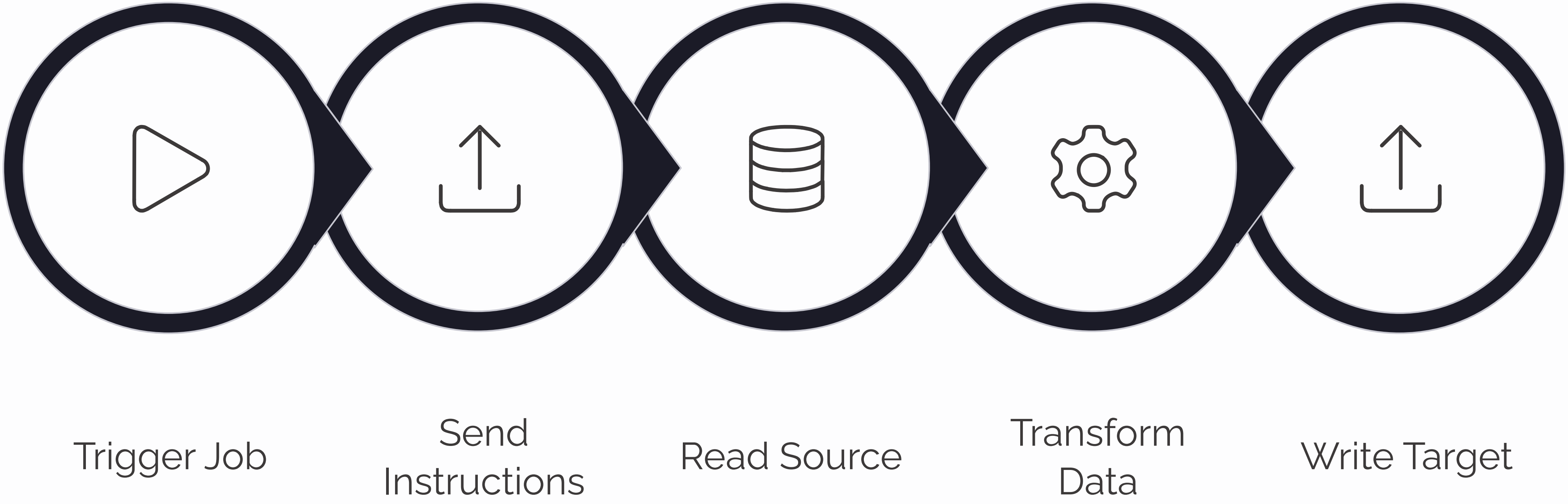


Secure Agent Runtime

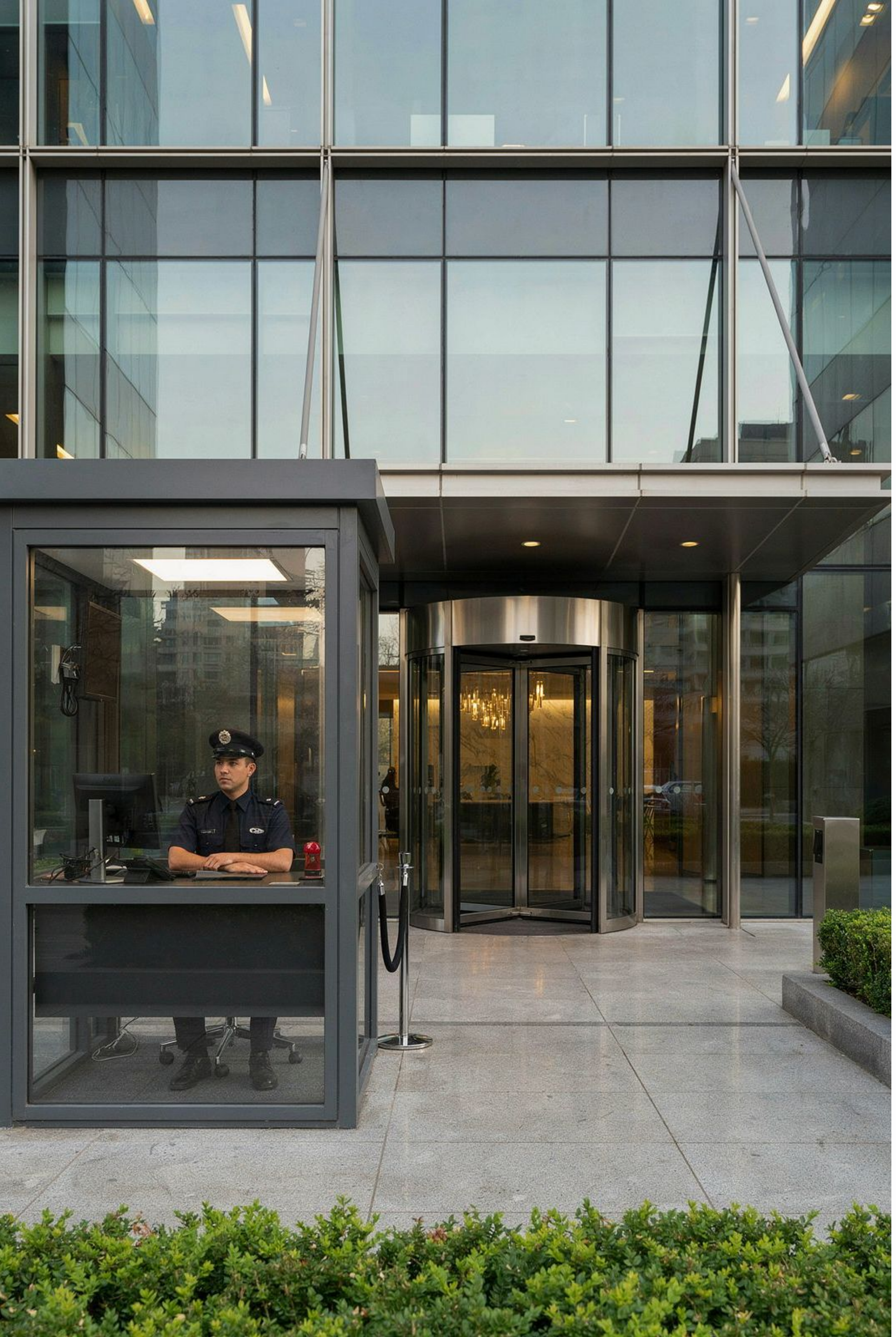
Your own server runs the job. Required when connecting to **on-premises databases**, internal file shares, or systems behind your firewall. The data flows through your network — Informatica only sends the job instructions, not your data.

Job Execution Flow — Step by Step

Here's what happens from the moment you click "Run" to when data arrives at the target. Knowing this flow helps you pinpoint exactly where to look when something goes wrong.



Every integration job follows this exact sequence. Notice that steps 3–5 all happen **inside your network** when using a Secure Agent — Informatica's cloud never touches your raw data.



Session 3: Secure Agent Fundamentals

The Secure Agent is your on-premises representative inside the IICS ecosystem. Think of it as a **trusted courier** who lives inside your building and can access all your internal systems — while still taking instructions from the cloud.

Secure Agent: Groups, Services & Status

A single agent is rarely enough for enterprise workloads. Here's how agents are organized and what they actually run.

Agent Groups

Multiple Secure Agents can be placed in a **group** for load balancing and high availability. If one agent is busy or down, another in the group picks up the job — like having multiple checkout lanes at a grocery store.

Agent Services

Each agent runs sub-services: **Data Integration Server** (runs mappings), **Mass Ingestion**, **Application Integration**, and more. Each service can be independently started, stopped, or monitored.

Checking Agent Status

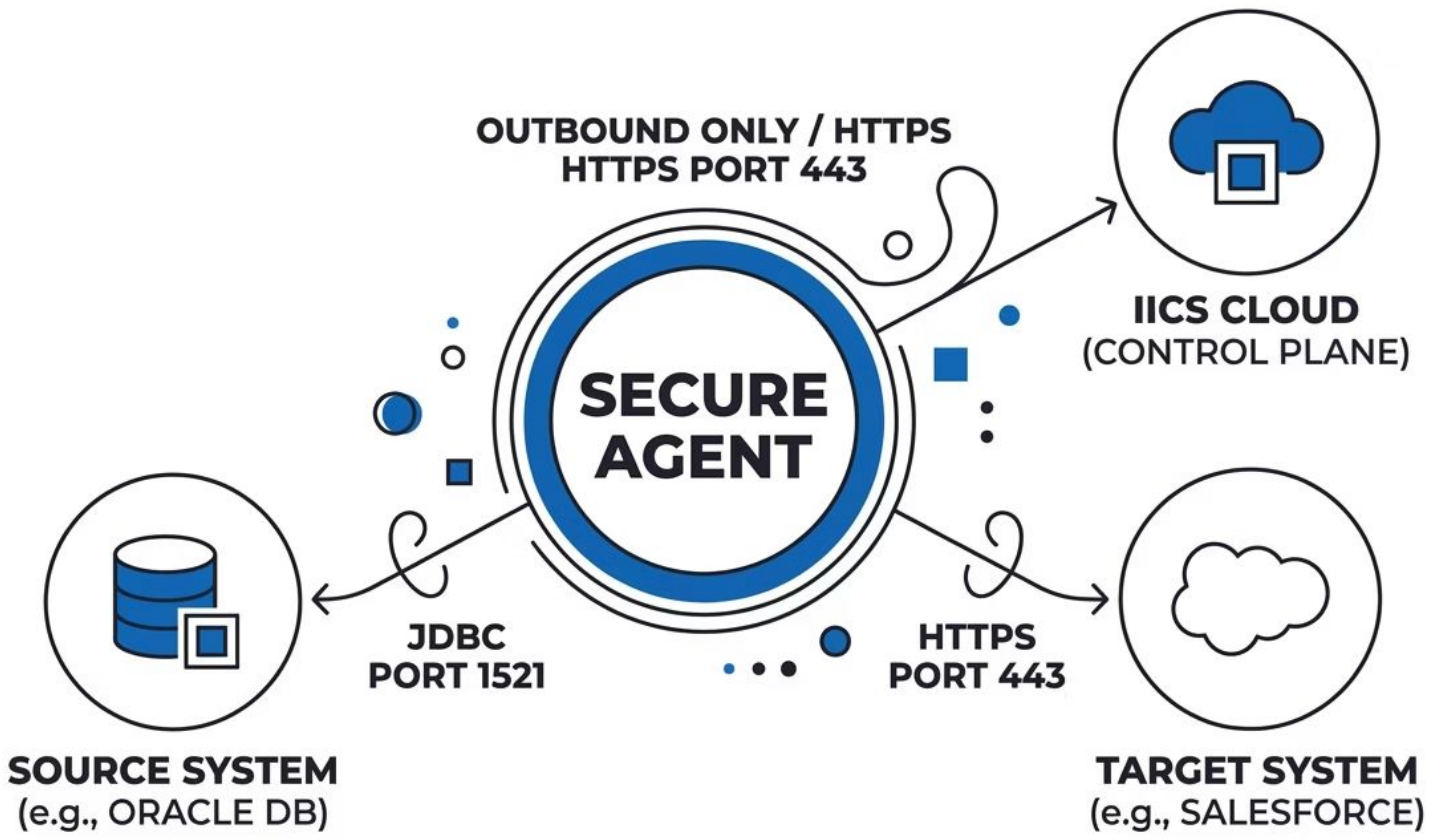
In the IICS UI, navigate to **Administrator → Runtime Environments**. Green = healthy. Yellow = degraded. Red = offline. Always check agent status first before running a job or debugging a failure.

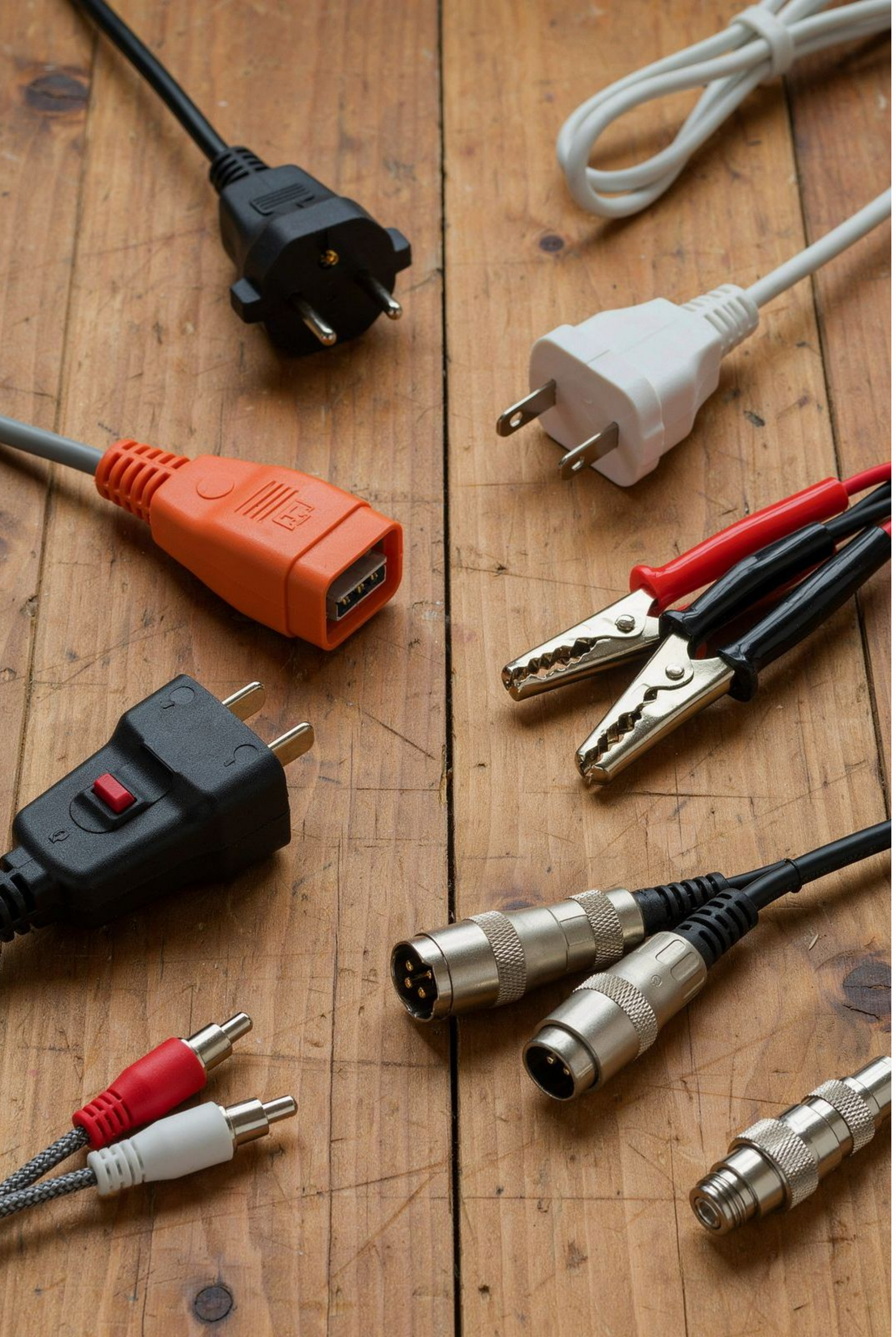
Basic Configuration

Agent settings live in `infaagent.ini` and the `conf/` directory. Key parameters include the agent proxy settings, max memory allocation, and the IICS org token used for authentication.

How IICS, Secure Agent & Systems Talk to Each Other

Connectivity is **outbound only** from the Secure Agent — it initiates the connection to the IICS cloud (port 443/HTTPS). Your firewall never needs inbound rules for IICS. This is a common source of confusion for network teams.





Session 4: Connections & Connectors

People often use "connector" and "connection" interchangeably — but in IICS, they mean **very different things**. Getting this distinction right will save you hours of confusion.

Connector vs. Connection: A Simple Analogy

A **Connector** is the *type* of plug — it defines what kind of system IICS can talk to and how. A **Connection** is a *specific configured instance* of that connector with real credentials and settings. You install a connector once; you create many connections from it.

🔌 Connector

The **driver and protocol definition**. Example: "Salesforce Connector" knows how to speak the Salesforce API language. It's like the USB-C standard — the specification itself, not a specific cable.

🔗 Connection

A **specific configured instance**. Example: "Salesforce_PROD" with username, password, security token, and environment URL filled in. It's the actual cable you plug in to connect to *that specific* Salesforce org.

📋 Connection Properties

Each connection has properties: hostname, port, database name, username, password, schema, SSL toggle, etc. These vary by connector type. Always consult Informatica docs for the required vs. optional fields.

🔍 Metadata Access

Once a connection is saved and tested, IICS can browse it — listing available tables, files, or API objects. This metadata is what populates your source/target selectors in the mapping designer.

Always **test your connection** immediately after creating it. A green test result confirms credentials, network path, and driver are all working before you build anything on top of it.

Session 5: Common Enterprise Connectors

IICS ships with hundreds of connectors. Here are the ones you'll encounter most often in enterprise environments — and when to reach for each one.

Flat File

CSV, delimited, fixed-width files on disk or shared drives. The universal fallback — almost every system can export a flat file. Used for batch loads, legacy system integration, and simple staging.

Relational Databases

Native connectors for Oracle, SQL Server, PostgreSQL, MySQL, and more. Use these when a native connector exists — they're faster and more feature-rich than generic JDBC.

JDBC / ODBC

Generic database connectivity for any JDBC/ODBC-compliant database that lacks a native connector. Requires placing the vendor's `.jar` driver file in the agent's `ext/` directory.

Salesforce

Purpose-built connector using Salesforce's Bulk API and REST API. Handles standard and custom objects, metadata, and change data. Best choice for any Salesforce integration.

REST

Connect to any REST API — internal microservices, third-party SaaS, or partner APIs. You define the endpoint, authentication method (OAuth, API key, Basic), and request/response schema.

SFTP & Cloud Storage

SFTP for secure file exchange with partners. Amazon S3, Azure Blob, and Google Cloud Storage connectors for modern data lake scenarios. Essential for both legacy and cloud-native pipelines.

Session 6: Connection Configuration Considerations

Getting a connection to work isn't just about entering a username and password. Real enterprise environments have firewalls, proxies, SSL certificates, and strict access controls. Here's what to check — and in what order.

1 Credentials & Permissions

Use a **dedicated service account** — not a personal user account. Ensure it has the minimum required permissions on the source/target (read-only on sources, write on targets). Shared credentials cause audit and rotation headaches.

2 Hostnames, Ports & Paths

Confirm the exact hostname or IP, port number, and database/schema/file path. A wrong port (e.g., 1521 vs. 1522 for Oracle) silently fails with a "connection refused" error that looks like a credential issue.

3 Drivers

For JDBC connections, copy the vendor's `.jar` file to the Secure Agent's `ext/` directory and **restart the Data Integration service**. The driver version must match the database server version.

4 Network, Firewall, SSL & Proxy

Confirm the Secure Agent host can reach the target system on the required port. If SSL/TLS is required, import the server certificate into the agent's truststore. If your network uses a proxy, configure it in the agent's settings file before testing any external connections.

Session 7: Metadata & Data Preview

Before you write a single transformation, always **validate your access** by browsing metadata and previewing data directly in IICS. This saves hours of debugging downstream.

What You Can Browse

- **Relational DB:** schemas → tables → columns with data types
- **Flat Files:** directory listings → file names → field delimiters
- **Salesforce:** standard and custom objects → fields → picklist values
- **Cloud Storage:** buckets/containers → folder paths → file listings

Why It Matters

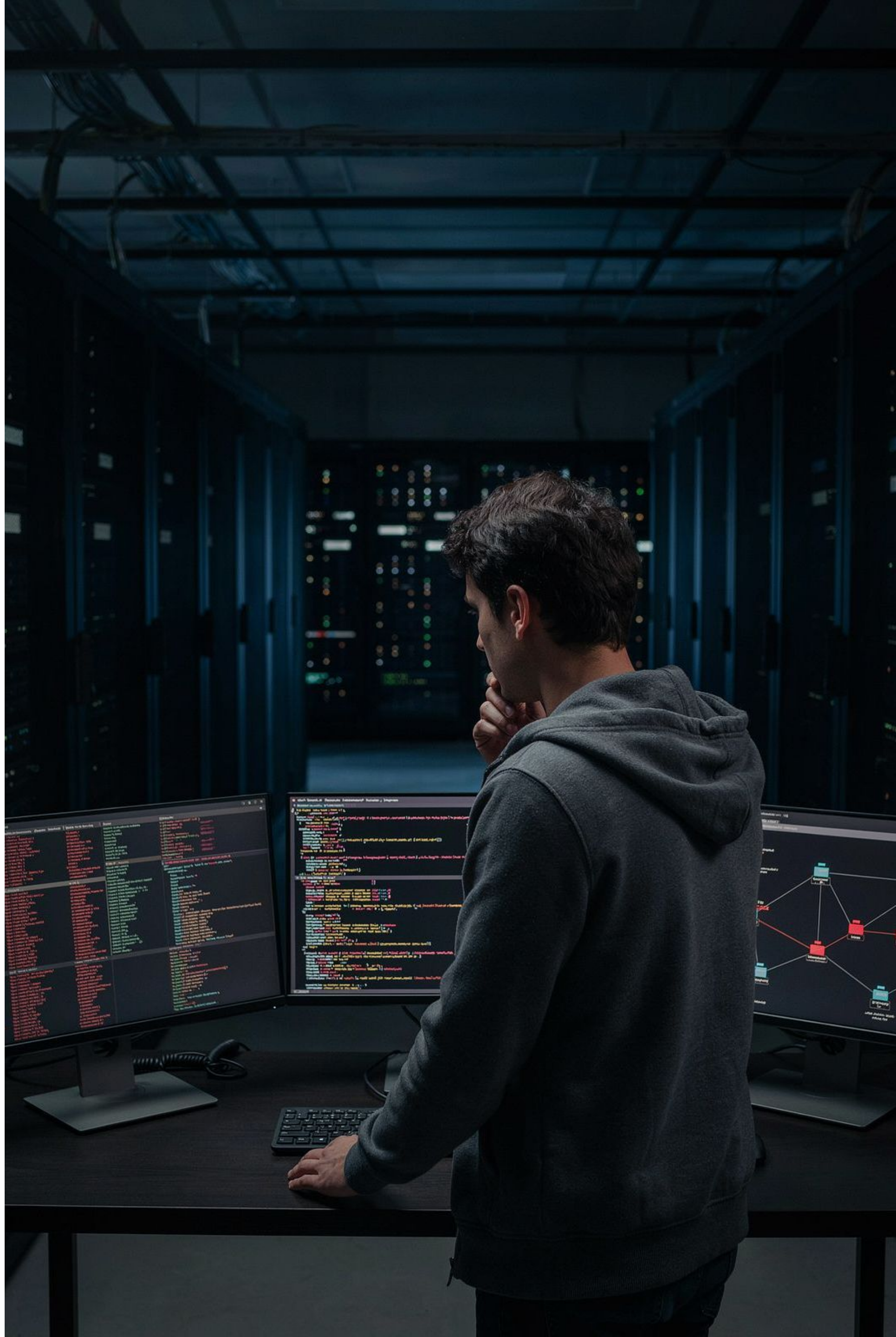
The **Source Preview** feature in the mapping designer pulls a live sample (usually 10–50 rows) directly from the source. Use it to:

- Confirm the connection account has **read access**
- Verify field names, data types, and sample values match expectations
- Catch encoding issues (e.g., special characters in names)
- Validate that filters or parameters will work before running a full job

Pro tip: If metadata browsing fails but the connection test passes, the service account likely has **connect privilege but not SELECT/read privilege** on the specific object. Check object-level permissions separately.

Session 8: Connector Troubleshooting

Things will go wrong. Knowing *where to look* and *what to check first* is the skill that separates experienced integration engineers from beginners. Let's build that instinct.



Common Failures & How to Fix Them

Use this as your **first-response checklist** when a connection test or job fails. Work top to bottom — most issues fall in the first three rows.

Failure Type	Likely Cause	Where to Look / Fix
Connection test fails	Wrong credentials, host unreachable, wrong port	Recheck host/port/credentials; ping from Agent host; check firewall rules
Secure Agent unavailable	Agent service stopped, server rebooted, memory exhausted	Check agent status in UI; SSH to agent host; restart Data Integration service
Database driver error	Missing or mismatched JDBC `.jar` file	Verify `.jar` in `ext/` dir; check driver version matches DB version; restart service
Authentication failure	Expired password, locked account, OAuth token expired	Reset service account password; refresh OAuth token; check account lock status in target system
File path / permission error	Wrong path, agent OS user lacks read/write access	Verify path exists from Agent host; check OS-level file/folder permissions for agent service user

Log locations: Agent logs are found at `[Agent Install Dir]/apps/Data_Integration_Server/logs/`. The `node.log` and `session logs` are your first stop. In the IICS UI, check **Monitor → Jobs** for session-level error messages with row-level detail.

Day 1 Key Takeaways

You've covered a lot of ground today. Here's what to carry forward into Day 2 — the mental models that will anchor everything else you learn.

Two-Plane Architecture

Cloud = orchestration. Secure Agent = execution. Your data stays on your network when using an agent.

Agent = Your On-Site Proxy

Outbound-only connections. No inbound firewall rules needed. Groups provide HA and load balancing.

Connector ≠ Connection

Connector is the driver type. Connection is the specific configured instance. Always test before building.

Validate Before You Build

Browse metadata and preview source data before designing any mapping. Catch access issues early.

Troubleshooting Playbook

Check agent status → verify credentials → confirm network/port → check driver → read the session log.

Tomorrow: We go deeper into **Mappings, Transformations, and Tasks**. Make sure your Secure Agent is installed and your first test connection is green before Day 2 begins.